

ABRAHAM JOSEPH

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PROFESSIONAL SUMMARY

Final-year B.Tech Electronics and Communication Engineering student specializing in embedded systems, IoT, and robotics. Proven experience in firmware development (C/C++), microcontroller programming (Arduino, ESP32, Raspberry Pi Pico W, TIVA C), and end-to-end IoT product development. Recognized with national-level awards including Best Software Award (SAE India) and IEEE hackathon prizes. Active technical leader, trainer, and community contributor.

TECHNICAL SKILLS

Programming:	C, C++, Python (basic)
Microcontrollers:	Arduino Uno/Nano, ESP32, ESP8266, Raspberry Pi Pico W, TIVA C, SIM800L GSM
Tools & Software:	KiCad, AutoCAD, Fusion 360, Arduino IDE, VS Code, Git, 3D Printing (FDM)
Protocols:	I2C, SPI, UART, GSM, Wi-Fi (HTTP / MQTT)
Domains:	Embedded Systems, IoT, Robotics, Real-Time Systems, PCB Design, Smart Automation
Soft Skills:	Technical Leadership, Public Speaking, Workshop Facilitation, Cross-functional Collaboration

WORK EXPERIENCE

Internship Trainee – Embedded Systems & Robotics **TORCEDU, Kochi** Jul 2024 – Oct 2024

- Developed C++ firmware for real-time embedded systems, applying OOP principles and hardware abstraction layer design.
- Debugged hardware-software interface issues using UART serial monitors, oscilloscopes, and logic analyzers.
- Implemented and optimised control algorithms for microcontroller-based systems under strict timing constraints.
- Collaborated on embedded development workflows and maintained version-controlled codebases.

Internship Trainee – Robotics & IoT Applications **CSEED, Kerala** Jan 2024 – Jun 2024

- Designed and simulated production-ready robotics layouts and embedded control system schematics.
- Built scalable IoT systems integrating GSM (SIM800L) and Wi-Fi modules for remote sensor data acquisition.
- Developed real-time firmware for environmental monitoring with sensor interfacing (DHT11, PIR, LDR).
- Contributed to full product cycles from circuit design (KiCad) through to 3D-printed enclosures.

PROJECTS

Smart Home Energy Saving System *Arduino Uno, PIR, LDR, DHT11, C++*

- Engineered an Arduino Uno-based home automation prototype using PIR, LDR, and DHT11 sensors to automatically control lighting and appliances based on occupancy and ambient conditions.
- Implemented C++ conditional logic firmware to reduce standby energy consumption intelligently.
- Coordinated a 7-member team across hardware assembly, firmware development, and system testing phases.

GSM / Wi-Fi IoT Environmental Monitoring System *ESP32, SIM800L, MQTT*

- Designed and deployed sensor nodes using ESP32 and SIM800L GSM for real-time environmental data relay over MQTT.
- Integrated dual wireless layers (cellular + Wi-Fi) to ensure reliability in low-connectivity environments.

Autonomous Robotics Control Architecture *C++, Microcontrollers, AutoCAD, 3D Printing*

- Programmed modular C++ control firmware for autonomous robot navigation with sensor fusion and PID control loops.
- Designed enclosures and chassis in AutoCAD and fabricated physical prototypes via FDM 3D printing.

EDUCATION

B.Tech – Electronics & Communication Engineering Christ College of Engineering (KTU), Irinjalakuda

Oct 2023 – Present

Relevant Coursework: Embedded Systems · Digital Signal Processing · Microprocessors & Microcontrollers · Communication Systems · IoT Architecture

CERTIFICATIONS

- Embedded Systems – NPTEL National Certification (Qualified & Certified)
- Industry Immersion in Embedded Systems – Digital Academy Certification Framework

AWARDS & ACHIEVEMENTS

- Best Software Award – SAE India (National-level technical development recognition)
- 1st Runner-Up + ₹30,000 Prize – IEEE QBit Hackathon (innovative embedded/IoT problem-solving)
- 2nd Runner-Up – Google TensorFlow Hackathon (ML/DL integrated software solution)

LEADERSHIP & VOLUNTEER EXPERIENCE

Technical Head ECTA – Christ College of Engineering

Nov 2023 – Oct 2024

- Led a 10+ member student technical committee to plan and execute national tech-fests, hackathons, and hands-on workshops.
- Coordinated logistics, faculty liaison, and content delivery for events serving 200+ participants.

Accessibility Co-Chair Young Indians (Yi)

Mar 2024 – Present

- Conducted campus accessibility audits and developed structured awareness programs across departments.

Community Co-Lead META HUB

Jul 2024 – Present

- Organised inter-collegiate engineering meets, open-source hardware sprints, and peer design reviews.

Volunteer IEDC Kerala

Aug 2024 – Present

- Supported regional startup conventions, entrepreneurship pitch panels, and technical ideation events.

KNOWLEDGE SHARING & TECHNICAL TRAINING

- Delivered workshops as a Resource Person across 7+ engineering institutions in Kerala.
- Trained 100+ engineering students per session in Robotics, IoT, Embedded Systems, and 3D Printing.
- Authored technical content for Tomson Electronics and published engineering articles on Medium.